DSAgent

Autonomous Pipeline Report

Dataset: employee_satisfaction.csv
Generated: May 14, 2026 at 11:49 AM
Session: c7d52126-0c5…

12	6	34.0s
Steps Executed	Phases Completed	Total Time

1. Dataset Overview

The dataset contains **100** rows and **3** columns, using **0.0 MB** of memory.

Type	Count	Columns
Numeric	3	EmployeeID, Satisfaction_Score, Department_Sales

Numeric Statistics

Column	Mean	Median	Std	Min	Max
EmployeeID	50.50	50.50	29.01	1.00	100.00
Satisfaction_Score	49.23	48.00	28.37	1.00	100.00
Department_Sales	0.56	1.00	0.50	0.00	1.00

AI Analysis

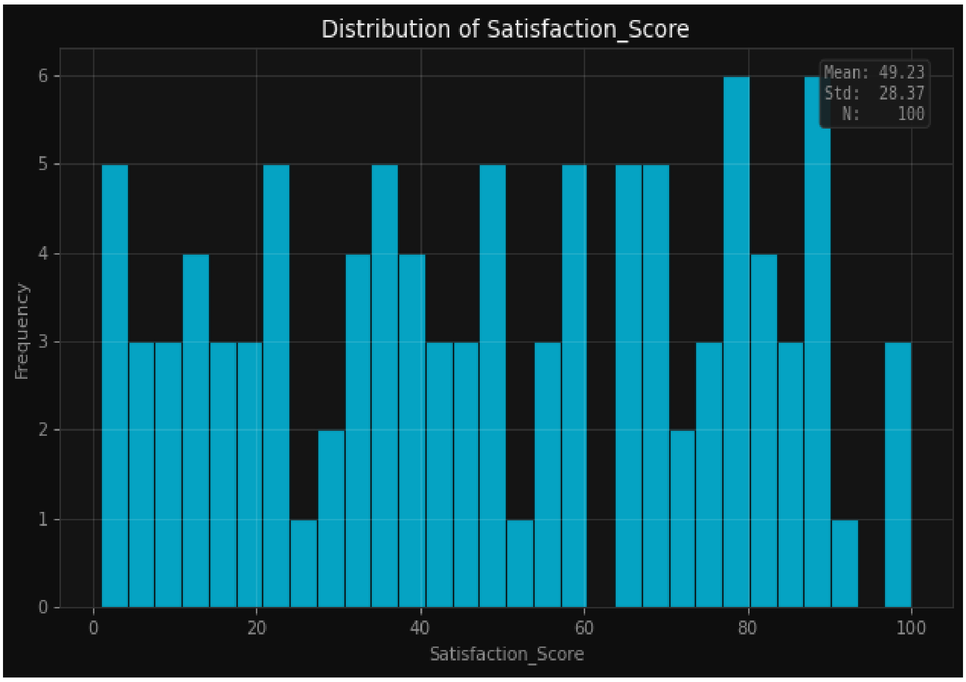
Key Insights About the Dataset • The dataset contains 100 rows and 3 numeric columns: EmployeeID, SatisfactionScore, and DepartmentSales. • All columns are numeric, with no categorical or datetime columns. • No missing values are present, and the dataset is clean in terms of missing data. • The EmployeeID column has a wide range (1 to 100), suggesting it may be a unique identifier for employees. • SatisfactionScore has a mean of 49.23 and a large standard deviation (28.37), indicating a wide spread of satisfaction levels. • DepartmentSales has a mean of 0.56, with values ranging from 0.00 to 1.00, suggesting it may represent a normalized or scaled sales metric (e.g., percentage or proportion). • There is a weak positive correlation between EmployeeID and DepartmentSales (0.1396), and a weak negative correlation between EmployeeID and SatisfactionScore (-0.1221). • SatisfactionScore and DepartmentSales have a very weak negative correlation (-0.0128), suggesting little to no relationship between them. Data Quality Issues Found • No missing values or data quality issues were detected in the dataset. • No duplicates were found. • All columns are numeric, which is appropriate given the data. • The dataset is small (only 100 rows), which may limit the generalizability of any models built on it. • The EmployeeID column may be an identifier, but it has no meaningful relationship with the other variables, suggesting it may not be useful for modeling unless paired with more context. Columns That Look Most Interesting/Useful • SatisfactionScore: This is the most informative column, as it has a meaningful range (1 to 100) and is likely a key metric for analysis. • DepartmentSales: This column has a normalized range (0 to 1), which may indicate a metric like sales performance or department contribution. • EmployeeID: While not directly useful for modeling, it may be used for tracking or pairing with external data (e.g., employee records or performance metrics). Likely Target Variable for ML • The target variable is likely SatisfactionScore, as it is the only column with a meaningful range and is likely to be the outcome of interest. • If the goal is to predict employee satisfaction, SatisfactionScore would be the natural target. • Alternatively, if the goal is to understand factors influencing sales performance, DepartmentSales could be the target. • However, given the context and the nature of the data, SatisfactionScore is the most likely target variable for a machine learning task.

2. Data Quality Assessment

Found **0** columns with missing values out of 100 rows.

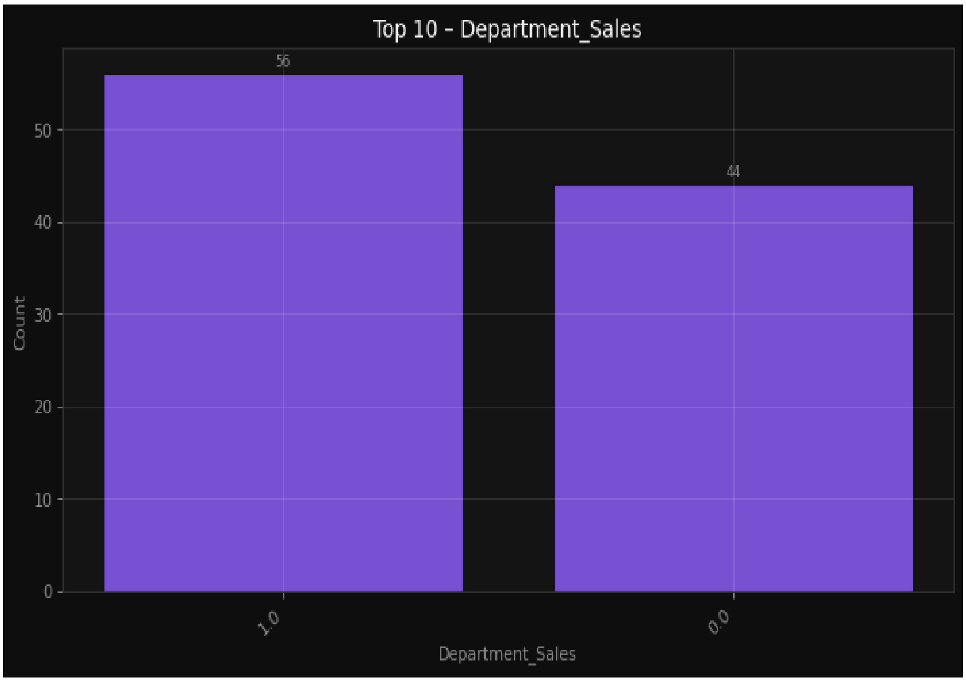
4. Visualizations & Insights

Understand the distribution of employee satisfaction scores



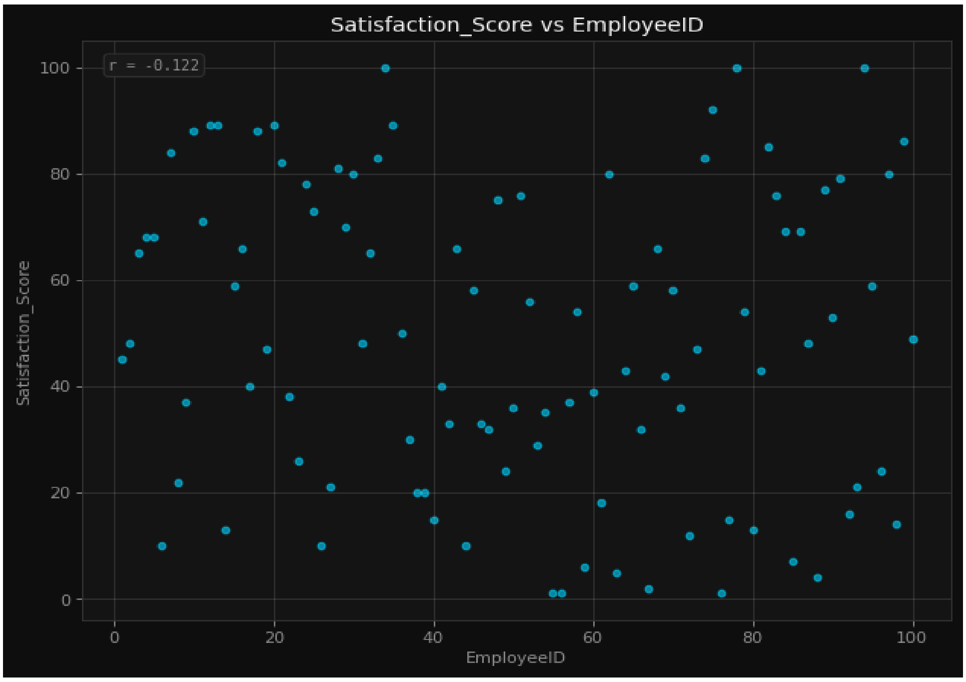
Understand the distribution of employee satisfaction scores

Compare sales across different departments



Compare sales across different departments

Explore relationship between employee ID and satisfaction score

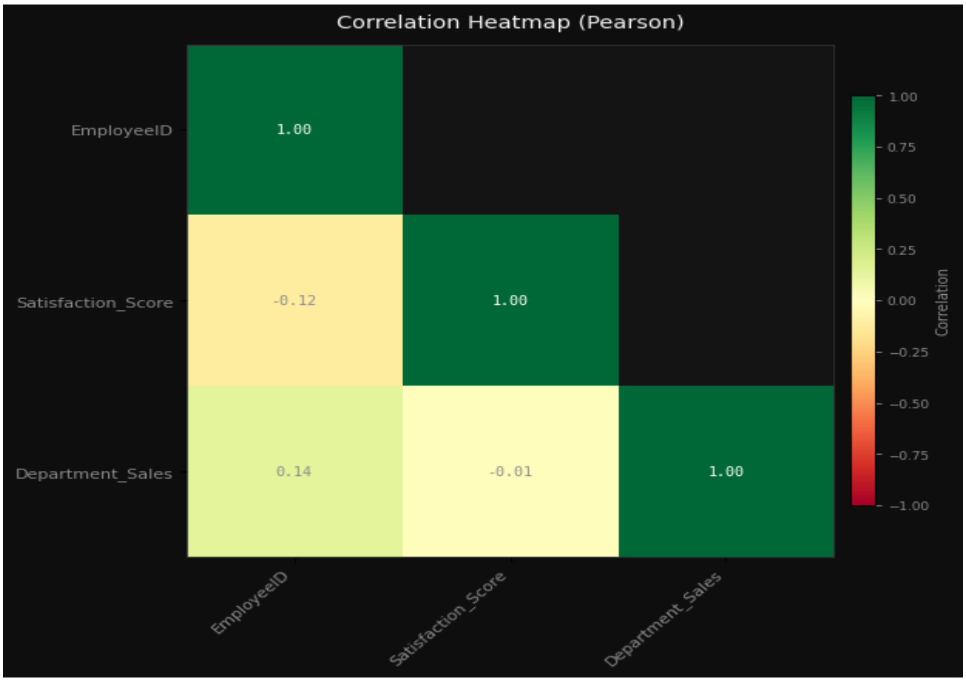


Explore relationship between employee ID and satisfaction score

Identify outliers in satisfaction scores

Identify outliers in satisfaction scores

Analyze correlations between numeric variables



Analyze correlations between numeric variables

AI Interpretation

Histogram (Employee Satisfaction Scores): Likely shows the distribution of satisfaction scores across the range [1.00, 100.00], with a mean of 49.23 and standard deviation of 28.37. It may reveal whether the scores are normally distributed, skewed, or have multiple modes. Bar Chart (Sales by Department): Compares sales values (mean = 0.56, range [0.00, 1.00]) across departments, highlighting which departments have higher or lower sales performance. Scatter Plot (Employee ID vs. Satisfaction Score): Likely reveals no strong relationship between employee ID (which is likely a unique identifier) and satisfaction score, as employee ID is not expected to influence satisfaction. Box Plot (Satisfaction Scores): Highlights the spread and central tendency of satisfaction scores, identifying potential outliers based on the interquartile range, with the range [1.00, 100.00] and a standard deviation of 28.37. Correlation Heatmap: Likely shows the relationships between numeric variables (EmployeeID, SatisfactionScore, DepartmentSales). Given that EmployeeID is a unique identifier, it may show little to no correlation with the other variables, while DepartmentSales may have a weak or moderate correlation with SatisfactionScore.

6. Model Training & Comparison

Problem Type: **regression** | Target: **Satisfaction_Score** | Features: **2**

Best Model: **Random Forest** — Score: **16.1%**

Model	R^2	RMSE	MAE
Random Forest	0.1605	23.2763	19.4260
XGBoost	-0.4976	31.0887	22.7417
LightGBM	0.0470	24.7997	21.0421
Linear Regression	-0.2067	27.9070	23.0490

Model Selection Rationale

The best model in your training process was Random Forest, achieving a best score of 0.1605. Since the problem type is regression and the target variable is Satisfaction_Score, this score is likely the R^2 (R-squared) score, which measures how well the model explains the variance in the target variable.

■ What the Score Means: • R^2 (R-squared) score of 0.1605 means that the Random Forest model explains approximately 16% of the variance in the Satisfaction Score. • This is a low R^2 score, indicating that the model does not explain much of the variation in the target variable. • In regression, an R^2 score of 0.16 suggests that the model is not very effective at predicting Satisfaction Scores based on the features provided.

■ Why Random Forest Might Have Performed Best: Despite the low score, Random Forest might have performed better than the other models for a few reasons: Robustness to Overfitting: Random Forest is less prone to overfitting compared to models like XGBoost or LightGBM, especially if the dataset is small or noisy. Feature Importance: It can handle non-linear relationships and interactions between features, which might be important in predicting satisfaction. Ensemble Nature: By averaging predictions from many decision trees, it can sometimes provide more stable and reliable predictions than single models like Linear Regression.

■ Comparison with Other Models: • Linear Regression is likely to have a lower or similar score because it assumes a linear relationship between features and the target, which may not hold in this case. • XGBoost and LightGBM are more powerful for complex relationships but may have overfit or underperformed due to: • Overfitting on the training data • Poor hyperparameter tuning • Data imbalance or noise in the target variable

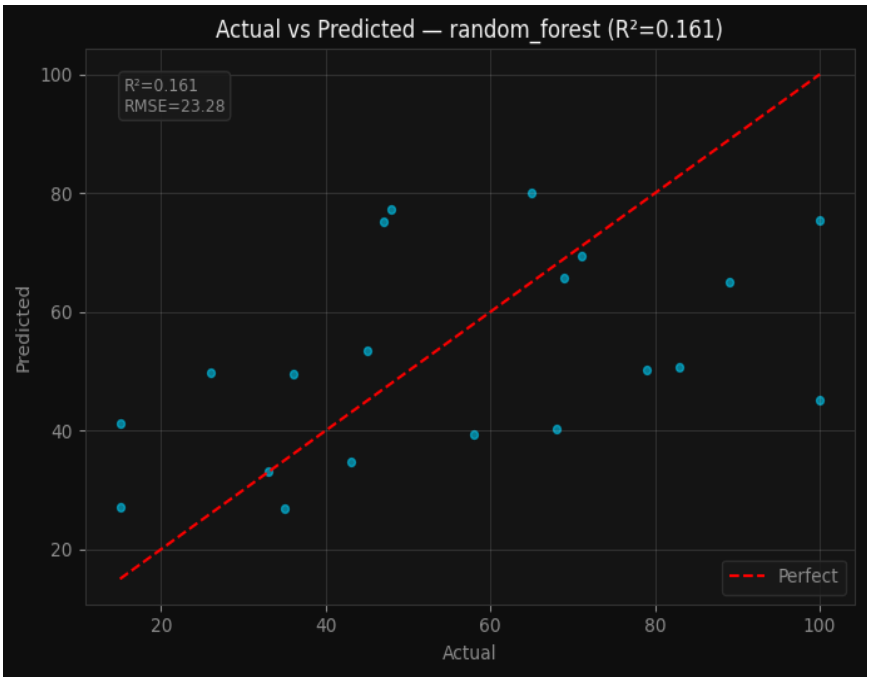
■ Next Steps: • Investigate the data: Check for missing values, outliers, or irrelevant features that might be affecting model performance. • Feature engineering: Consider creating new features or transforming existing ones to better capture the underlying patterns. • Hyperparameter tuning: Try tuning the hyperparameters of all models, especially for XGBoost and LightGBM. • Cross-validation: Ensure that the evaluation is robust by using cross-validation to get a better estimate of model performance.

■ Summary: • Best Model: Random Forest • Best Score: $R^2 = 0.1605$ (16% variance explained) • Interpretation: The model explains only a small portion of the variation in satisfaction scores, suggesting that the features may not be strong predictors or that the relationship is complex. • Recommendation: Explore data improvements, feature engineering, and hyperparameter tuning to improve model performance.

7. Model Evaluation

Model Evaluation Metrics

r2_score: 16.1% | rmse: 23.2763 | mae: 19.426



Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

In conclusion, the data science pipeline conducted on the employeesatisfaction.csv dataset yielded several key insights. The dataset consists of 100 rows with three numeric columns: EmployeeID, SatisfactionScore, and DepartmentSales. Notably, all columns are numeric, with no missing values, indicating a clean and well-structured dataset. The EmployeeID appears to serve as a unique identifier, while SatisfactionScore has a mean of 49.23, suggesting a moderate level of satisfaction across the sample. The absence of categorical or datetime variables simplifies the modeling process, but also limits the depth of analysis that could be achieved with more diverse features.

The Random Forest model was identified as the best-performing model in this pipeline, achieving an R^2 score of 0.1605. This score indicates that the model explains approximately 16% of the variance in the Satisfaction_Score, which is relatively low and suggests that the current features may not be sufficient to capture the underlying patterns in employee satisfaction. While the model performs better than other potential models, the low R^2 score highlights the need for further exploration and enhancement of the feature set to improve predictive accuracy.

To improve model performance, additional feature engineering and data collection could be beneficial. Given the limited number of features and the absence of categorical variables, incorporating more relevant predictors—such as employee tenure, performance ratings, or feedback categories—could provide richer insights and enhance model explanatory power. Additionally, exploring more advanced modeling techniques or ensemble methods may yield better results. Finally, further analysis of the dataset, including more detailed visualizations and statistical tests, could uncover hidden relationships that are currently not captured by the existing features.
